INCH-POUND

MIL-S-83862A(USAF)
16 November 1990
SUPERSEDING
MIL-S-83862(USAF)
9 July 1973

MILITARY SPECIFICATION

STRAP, RETENTION, SURVIVAL KIT

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the retention strap assembly installed in rigid seat survival kits. The strap assembly restrains the inner container or rucksack in the survival kit upon ejection from an aircraft.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Boxes, Shipping Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Support Division, San Antonio ALC/TIRDM, Kelly AFB TX 78241-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1680

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MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-543	Buckle, Tongueless and Web Strap
MIL-W-4088	Webbing, Textile, Woven Nylon
MIL-T-43636	Thread, Nylon, Non-Melting

STANDARDS

FEDERAL

FED-STD-191	Textile Test Methods
FED-STD-751	Stitch, Seam, and Stitching

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-147	Palletized Unit Load
MIL-STD-970	Standards and Specifications, Order of Precedence For the Selection of
MIL-STD-831	Test Reports, Preparation of
MIL-STD-2073-1	DOD Materiel, Procedures for Development And Application of Packaging Requirements

(Unless otherwise indicated copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, BLDG 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in solicitation.

NATIONAL BUREAU OF STANDARDS

HQ DPSC Handbook 4120.1	List of Armed Services Colors for Clothing

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM-D-3951 Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AMS 3798/3 Webbing, Low Modulus Aramid

(Application for copies should be address to Society of Automotive Engineers, 400 Commenwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3 Materials. Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Nonmetallic materials. Any nonmetallic materials which corrode, deteriorate, or are affected adversely by extremes in atmospheric conditions shall not be used. The nonmetallic material shall retain its initial characteristics for prolonged periods without any signs of deterioration such as setting, weave separation, or fungus attack.

3.3.2 Adhesives. Any adhesives which are used shall resist attack by fungus, moisture, or other foreign environment which may come in contact with the strap.

3.3.3 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to occur in storage or normal service. Metals shall be protected against

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such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective treatment that will crack, chip or scale with age or extremes of atmospheric conditions shall be avoided.

3.4 Design and construction. The retention strap assembly shall consist of two straps fastened by a quick release buckle assembly suitable for activation with one hand. One strap shall consist of a D ring and plate attached to a lenght of webbing specified in Figure 1. The second strap shall consist of a D ring and plate, one-hand quick release fastener with tab, and lenght of webbing attached according to Figure 1. The strap assembly shall be complete when the two straps are fastened together by the buckle. Color requirements of the assembly are specified in accordance with HQ DPSC Handbook 4120.1.

3.4.1 D ring and plate. The D ring and plate shall be P/N 1954 manufactured by North and Judd Mfg. Co., New Britain, Conn. or equivalent.

3.4.2 Webbing. The strap material shall be that webbing which is specified in AMS 3798/3, Type I, Solution Dyed, olive green color number 106 per HQ DPSC Handbook 4120.1 of the required lenght as specified in Figure 1. The webbing shall be seared at the ends to prevent fraying.

3.4.3 Release tab. The release tab shall be webbing per MIL-W-4088: Type I, yellow color number 1365 per HQ DPSC Handbook 4120.1 and of the required lenght as specified in Figure 1.

3.4.4 Buckle. The buckle shall be a Type V, Size 3/4, Class 3, in accordance with MIL-B-543 or any other buckle approved by the qualifying activity. The buckle shall be modified or designed to provide a one hand quick release operation using the attached release tab. The buckle shall have a fastened height of not more than 0.75 inches and a fastened lenght of not more than 1.5 inches. The buckle shall be designed so as to enable the strap assembly to be adjustable to conform to irregular shapes.

3.4.5 Thread. The thread shall be nylon in accordance with MIL-T-43636, Type I, Size F, and shall be of the same color as the webbing.

3.4.6 Stitching. Stitching shall be type 301 in accordance with FED-STD-751. All approved stitch patterns such as the "W" and "Box" shown on Figure 1 may be used as Long as the stitching has the appropriate number of stitches per inch and the necessary lenght to provide the strength required for the pull test as specified (see 3.5.2 and 4.6.2.1). All stitching shall be 1/8 inch from the edge of the material.

3.5 Performance.

3.5.1 Buckle. The buckle shall hold without slipping and shall release the strap quickly when the release tab is pulled with one hand while a Load of 200 pounds is applied to the strap assembly.

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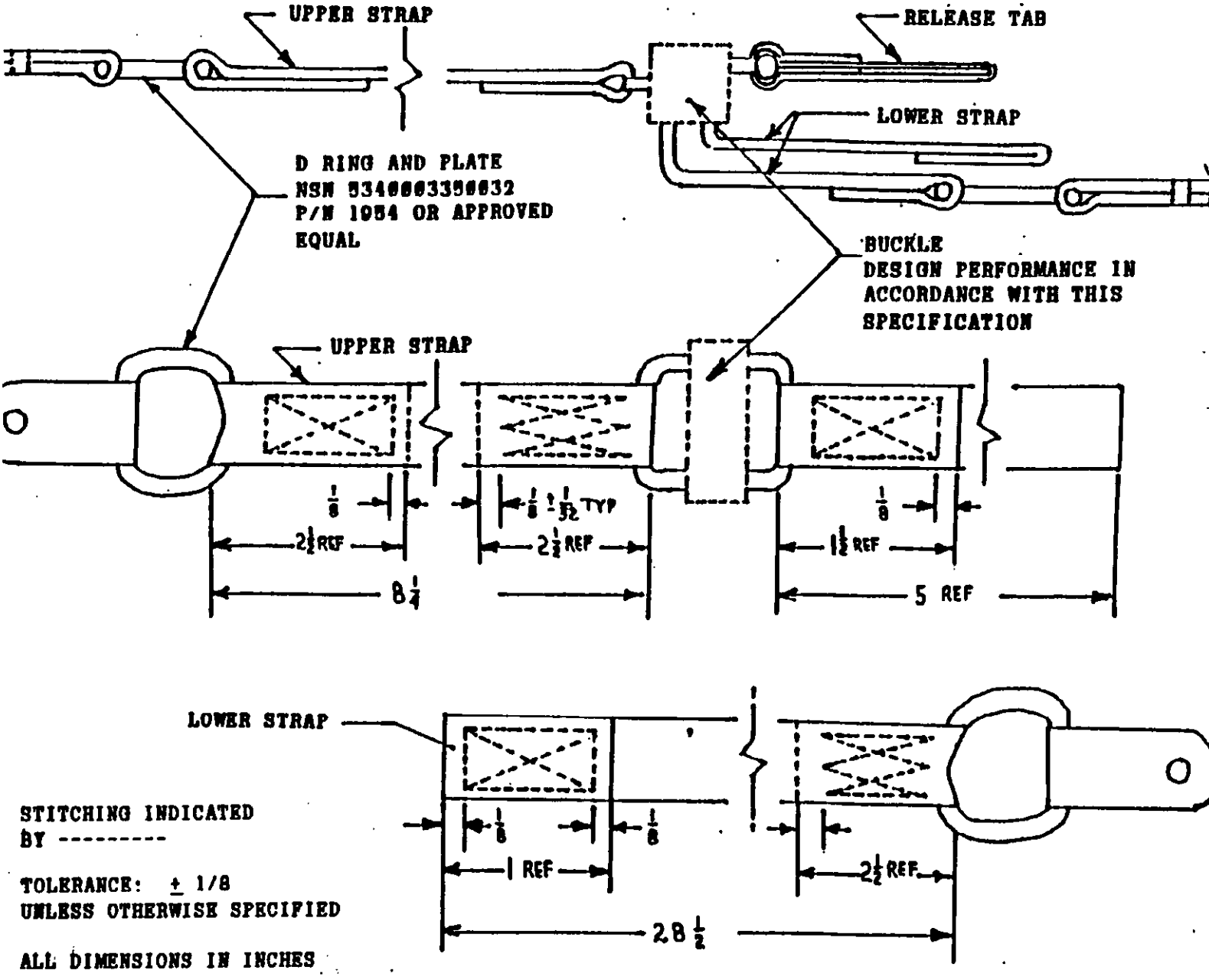


FIGURE 1. Retention strap assembly

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3.5.2 Strap. The strap shall be fastened securely to the D ring and buckle either by stitching or with a bonding adhesive which will permit strap to be pulled with a minimum of 400 lbs without failure.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-STD-100 shall govern changes in the manufacturer's part numbers.

3.7 Identification of product. The retention strap assembly shall be marked with the following information in accordance with MIL-STD-130. The retention strap assembly shall have the following information permanently affixed or impregnated on the release tab in lettering not less than 1/16 inch high and not more than 1/8 inch high:

Manufacturer's Code: *
Manufacturer's Part Number: *
Government Designation: MIL-S-83862
Contract Number: *

3.8 Workmanship. The retention strap assembly shall be uniform in quality and shall be free from irregularities, defects, or foreign matter which could adversely affect safety, performance, reliability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

* Manufacturer shall include the applicable information.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in FED-STD-191.

4.4 First article inspection. First article inspection shall be preformed on one complete retention strap assembly when a first article sample is required (see 3.1 and 6.2). This inspection shall include the examinations and tests as described herein (see 4.6).

4.4.1 First article. When specified in the contract, three complete first articles shall be submitted to the procuring activity at the time the first article test report is submitted for approval. The Government reserves the right to preform any or all tests, inspections and examinations specified herein.

4.4.2 Test reports. Three copies of a test report shall be furnished to the qualifying activity. The test report shall include the results of all tests and a detailed statement of compliance or noncompliance with each requirement of this specification, identified by the applicable paragraph number (see 6.3).

4.5 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.6.1 and the tests of 4.6.2.1 and 4.6.2.2.

4.5.1 Individual tests. The tests as specified (see 4.6.1) shall constitute individual tests.

4.5.2 Sampling tests. One retention strap assembly out of a Lot of 100, or fraction thereof, shall be subjected to the tests specified (see 4.6.2.1).

4.5.3 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.6 Methods of inspections.

4.6.1 Visual examination. Visual examination shall consist of the tests described in Table I. One major defect shall constitute rejection. Three minor defects out of a Lot of 25 shall constitute rejection.

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TABLE I	
CLASSIFICATION OF DEFECTS FOR VISUAL EXAMINATION	
MAJOR	MINOR
1. Assembly not complete. D Ring and Plate, Release, Tab, or Buckle missing. 2. Stitching loose or incomplete; or bonding adhesive gapped or missing. 3. Buckle damaged or use otherwise impaired. 4. Strap material cut or seriously frayed which will hinder intended use. 5. Dimensions not within tolerance limits of $^{1}/_{16}$ inch as specified in Figure 1. 6. D Ring and Plate damaged. 7. Any material which does not meet specifications.	101. Release tab or strap not of color specified. or Buckle missing. 102. Marking missing or illegible. 103. Straps not joined properly through buckle.

4.6.2 Performance.

4.6.2.1 Pull test. One retention strap assembly shall be tested per method 4108 of FED-STD-191 to measure buckle and bond strength of the strap assembly. The buckle or bond shall meet the requirements of 3.5.1 and 3.5.2.

4.6.2.2 Cyclic pull test. A cyclic pull test of 100 cycles shall be performed on one retention strap assemble. One D ring and plate shall be retained by a stationary jaw and the other D ring and plate shall be retained by a movable jaw of a suitable test device. A tensile load varying from 0 to 100 pounds shall be applied to the strap and buckle at a uniform pulling speed of 12 $\pm$ 1/2 inches per minute and maintained for a period of 5 seconds at 100 pounds before returning to zero. This will constitute one cycle. The test shall be started by pulling the webbing through the adjustor to eliminate any set in the webbing.

4.7 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable

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container specification shown in section 5. The inspection of marking for shipment and storage shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

5. PACKAGING

5.1. Preservation, packaging and packing. No content preservations shall be used. The two separate strap parts shall be fastened with the buckle, packed together, and shall constitute one strap assembly. Separate strap parts shall be shipped and packaged separately only at the request of the procuring activity. The retention strap assembly shall be packaged in accordance with Method III of MIL-P-116. Preservation shall be Level A, C or Commercial as specified (see 6.2).

5.1.1 Level A. Strap assemblies shall be preserved to the MIL-STD-2073-1 requirements for this Level.

5.1.2 Level C. Strap assemblies shall be preserved to the MIL-STD-2073-1 requirements for this level.

5.1.3 Commercial. Strap assemblies shall be preserved in accordance with the applicable requirements of ASTM-D-3951.

5.2 Packing. Packing shall be Level A, B, C, or Commercial as specified (see 6.2).

5.2.1 Level A. Strap assemblies that have been preserved and packaged as specified (see 5.1) shall be packed in exterior-type shipping containers that conform to PPP-B-601, overseas type. The closure of the shipping container shall be in accordance with the appendix of the applicable shipping container specification.

5.2.2 Level B. Strap assemblies that have been preserved and packaged as specified (see 5.1) shall be packed in exterior-type shipping containers that conform to PPP-B-636, class weather resistant. The closure of the shipping container shall be in accordance with the appendix of the applicable shipping container specification.

5.2.3 Level C. Strap assemblies that have been preserved and packaged as specified (see 5.1) shall be packed in accordance with the requirements of MIL-STD-2073-1.

5.2.4 Commercial. Strap assemblies that have been preserved and packaged as specified in (see 5.1) shall be packed in accordance with the requirements of ASTM-D-3951.

5.3 Marking

5.3.1 Levels A, B, and C. In addition to any special or other identification markings required by the contract (see 6.2), each unit pack and exterior container shall be marked in accordance with MIL-STD-129.

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5.3.2 Commercial. Strap assemblies shall be marked in accordance with the applicable requirements of ASTM-D-3951.

5.4 Palletization. Utilized loads, commensurate with the level of packing specified in the contract or order shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern shall first be approved by the contracting officer.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The retention strap assembly shall be used to restrain the inner container or rucksack of the survival kit when ejection from an aircraft is necessary. Since there is a chance that injury may occur to the crewmember prior to or on ejection, the strap assembly shall be capable of being released with one hand so as to enable the crewmember to readily remove the rucksack from the survival kit.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2)

c. List of interchangeable components and assemblies.

d. When first article is required (see 3.1, 4.4 and 4.4.1).

e. Levels of preservation and packing (see 5.1 and 5.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title
4.4.1	DI-T-5329	Inspection test report

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The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory , and the number of items to be tested as specified in 4.4. The contracting officer should also include specified instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Subject term (key word) listing.

Buckle
Reports, Test
Ring, D
Stitch, Seam
Thread, Nylon
Webbing

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Air Force - 99

Preparing activity:
Air Force - 82

Agent activity:
Air Force - 99

(Project 1660-F544)

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